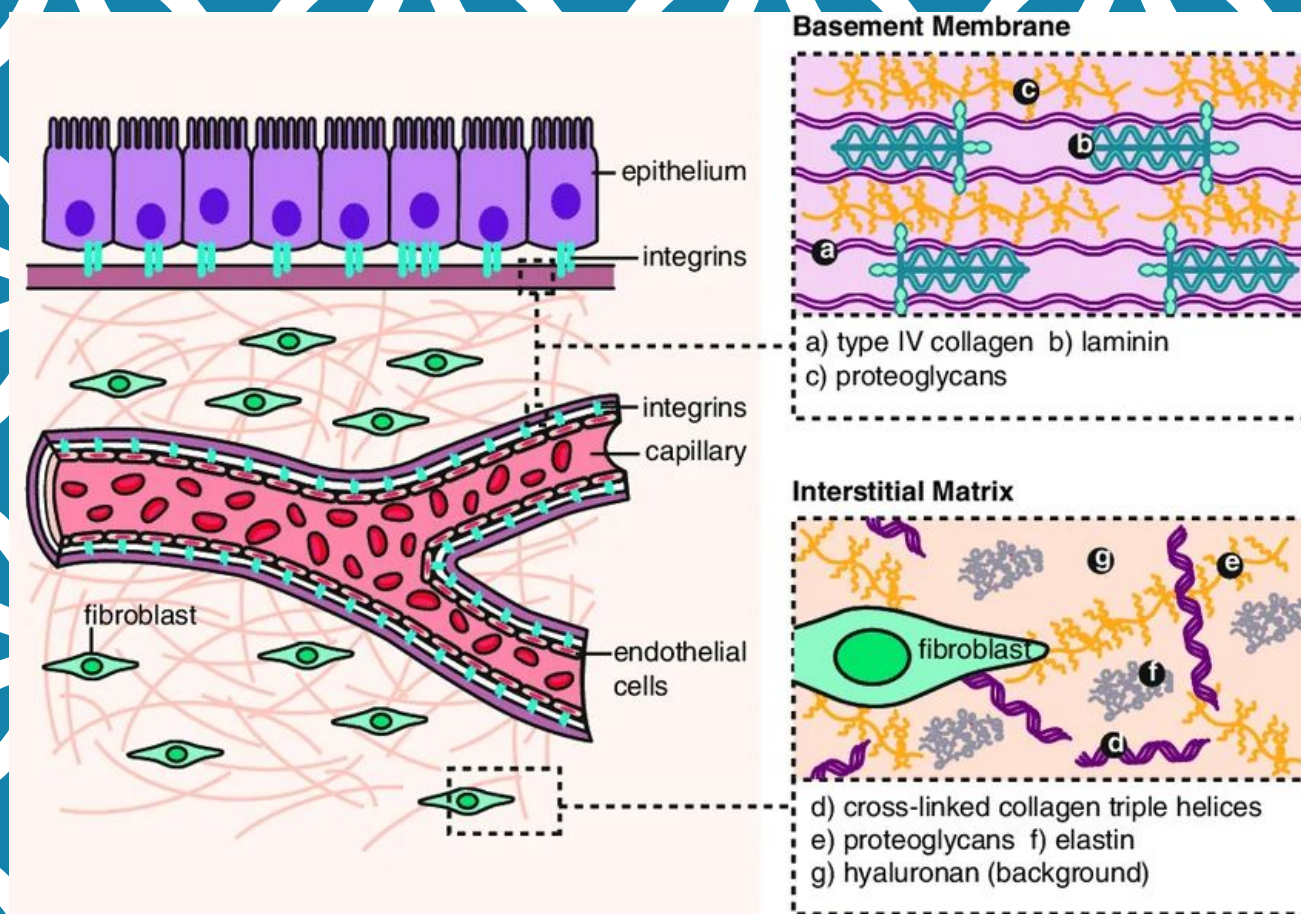




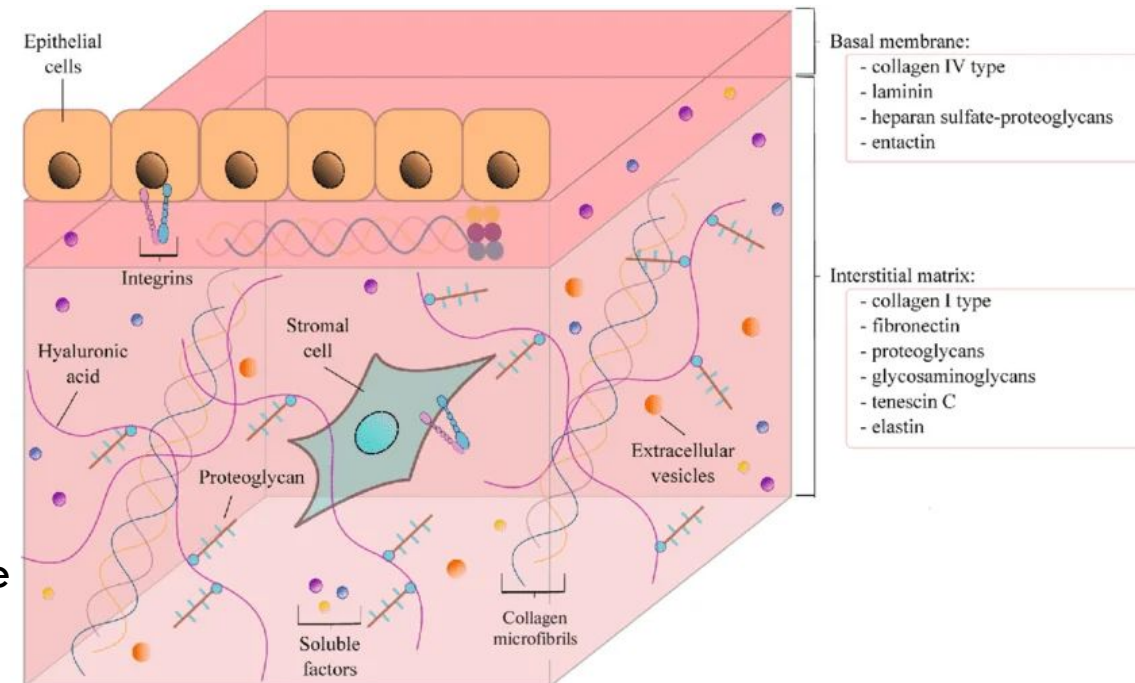
IMPORTANCE OF THE EXTRACELLULAR MATRIX IN HEALTH AND DISEASE

DR. WAJEEHA SHAMS ANSARI

INTRODUCTION: THE DYNAMIC MICROENVIRONMENT

The **Extracellular Matrix (ECM)** is not a static filler; it is a complex network of macromolecules that regulates cell behavior, including proliferation, survival, and differentiation. In biochemistry, we classify the ECM into two main forms:

- 1. Interstitial Matrix:** Found in the spaces between cells (connective tissue).
- 2. Basement Membrane:** A specialized, thin, tough sheet of ECM that separates epithelial cells from underlying connective tissue. Highly organized sheet (Type IV Collagen + Laminin) underlying epithelia.



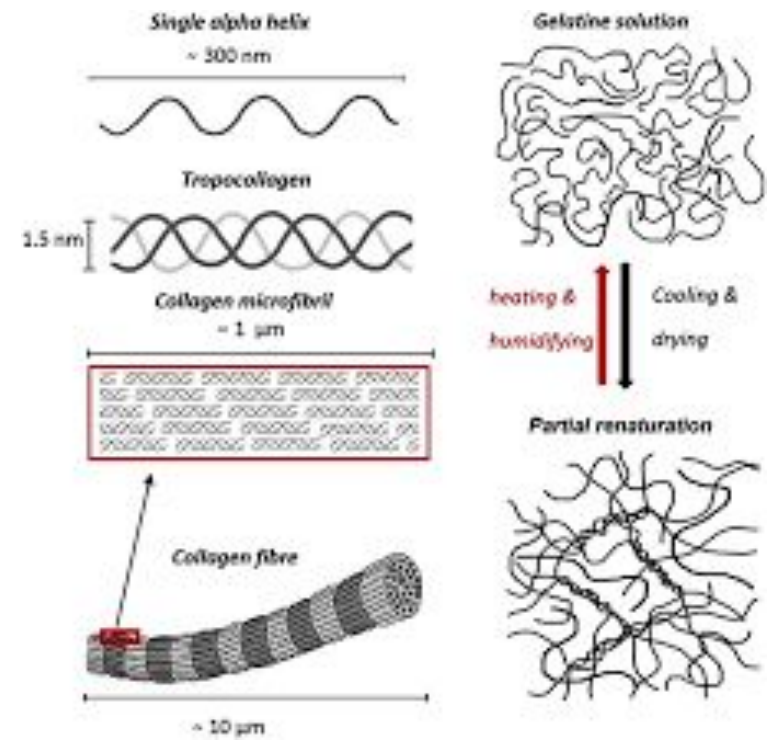
MOLECULAR COMPONENTS I – STRUCTURAL PROTEINS

- **Collagen (The Steel Cables):**

- The most abundant protein; triple helix structure.
- Requires Prolyl/Lysyl hydroxylase (Vitamin C dependent) for stability.

- **Elastin (The Rubber Band):**

- Rich in Glycine and Valine; contains **Desmosine** cross-links for recoil.



MOLECULAR COMPONENTS II – ADHESION & HYDRATION

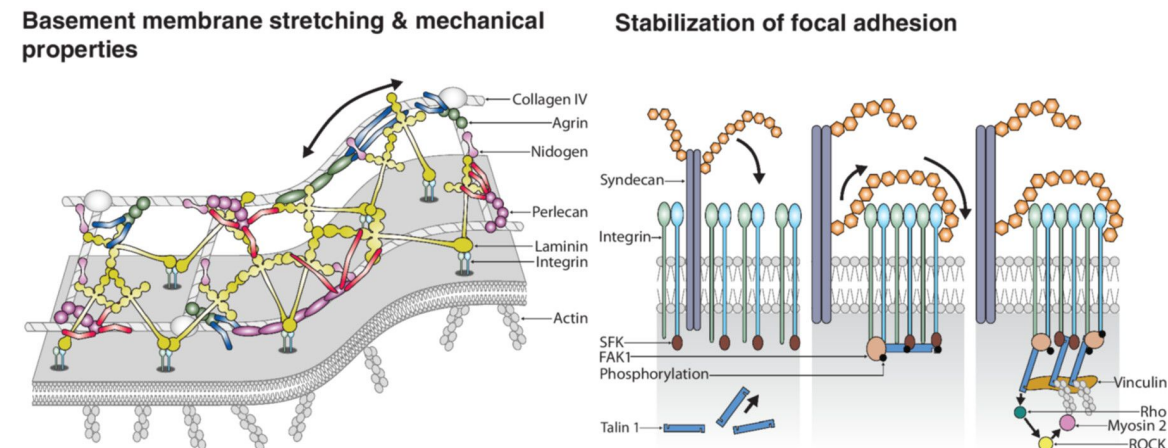
- **Adhesion Proteins:**

- **Fibronectin:** Connects cells to collagen via Integrins.

- **Laminin:** Primary organizer of the Basement Membrane.

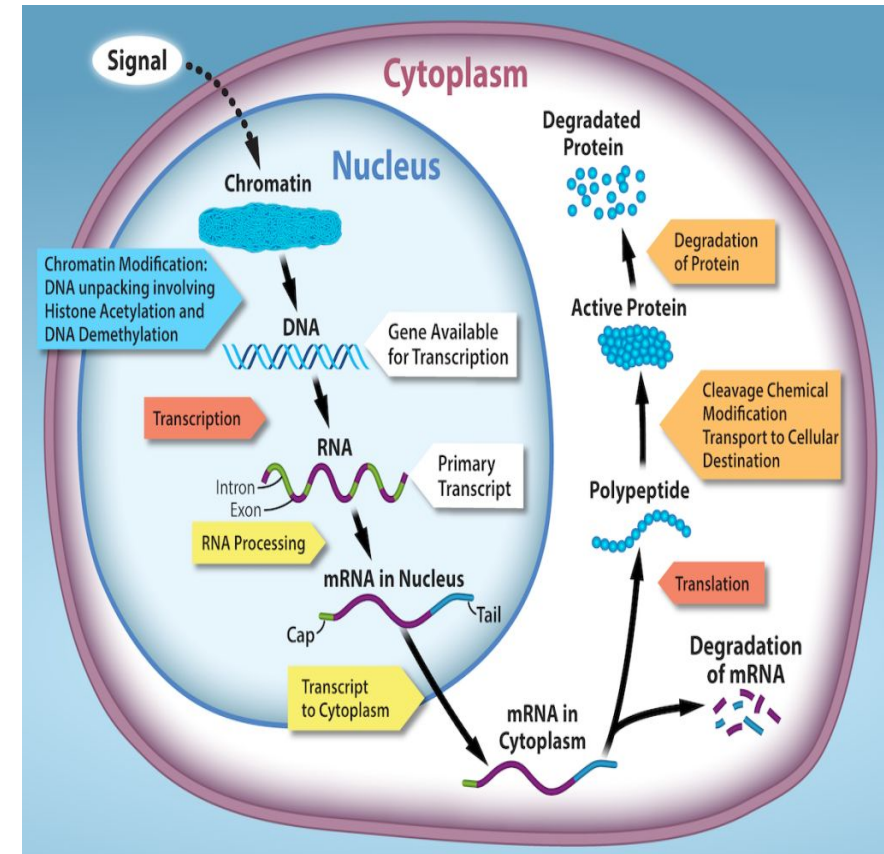
- **Ground Substance (GAGs & Proteoglycans):**

- Highly negative charge -> Attracts H₂O -> Creates a gel that resists compression.



IMPORTANCE IN HEALTH – PHYSIOLOGICAL ROLES

- **Mechanical Support**: Defines tissue shape and hardness (e.g. bone, etc).
- **The Signaling Hub**: Sequesters growth factors; regulates cell proliferation and survival.
- **Mechanotransduction**: How cells "sense" ECM stiffness through **Integrins**, converting physical force into biochemical signals.



ECM IN DISEASE I – GENETIC DEFECTS

- **Scurvy:** Defective collagen hydroxylation (Vitamin C deficiency).
- **Osteogenesis Imperfecta:** Mutations in Type I Collagen leading to "Brittle Bones."
- **Ehlers-Danlos Syndrome (EDS):** Faulty collagen processing/assembly; skin hyperextensibility.

ECM IN DISEASE II – ACQUIRED PATHOLOGIES

- **Fibrosis:**

- Excessive ECM deposition (e.g., Liver Cirrhosis, Pulmonary Fibrosis).

- **Cancer Metastasis:**

- Tumors secrete **Matrix Metalloproteinases (MMPs)**.
- MMPs degrade the basement membrane, allowing cancer cells to invade the bloodstream.

DISORDERS OF COLLAGEN

COLLAGENOPATHIES

Collagen disorders may be

Genetic

Autoimmune

Miscellaneous e.g. nutritional, drug-induced.

All these affect biosynthesis assembly, post-translational modification, secretion, or other process involved in normal collagen production.

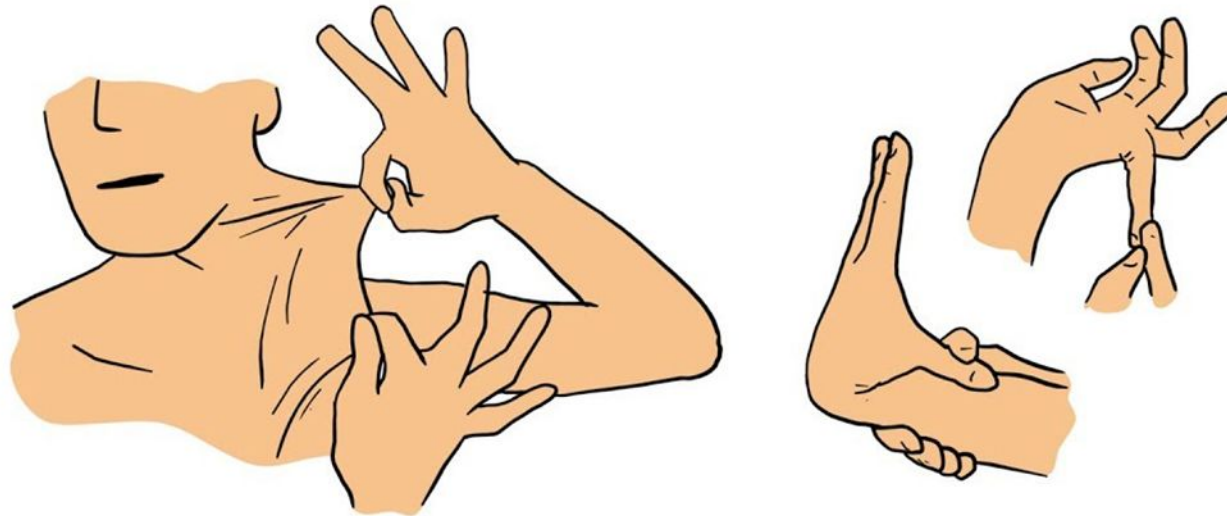
EHLERS-DANLOS SYNDROME

Ehlers-Danlos syndrome (EDS) is a heterogeneous group of **connective tissue disorders** that result from **inheritable defects** in the metabolism of fibrillar collagen molecules.

EDS can be caused by a **deficiency of collagen-processing enzymes** or from mutations in the amino acid sequences of collagen types I, III, or V.

The **classic form of EDS**, caused by defects in type V collagen, is characterized by skin extensibility and fragility and joint **hypermobility**.

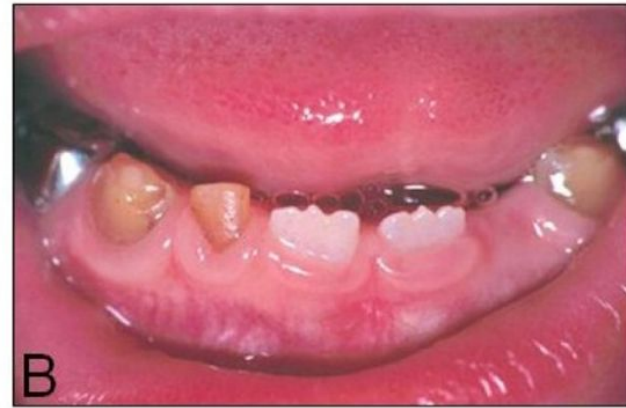
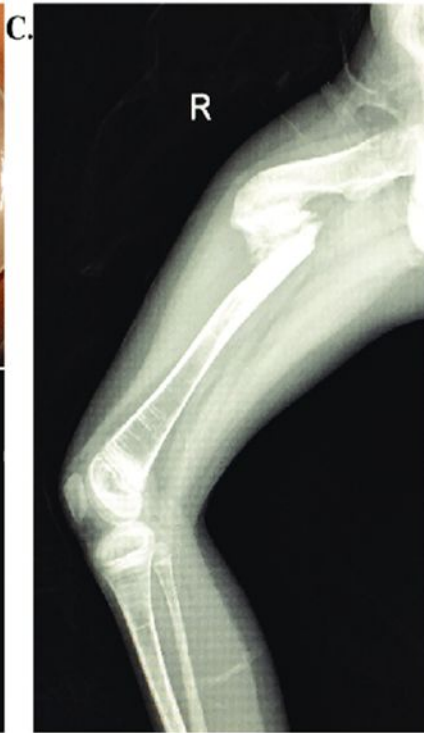
The vascular form, due to defects in type III collagen, is the most serious form of EDS because it is associated with potentially lethal arterial rupture.



OSTEOGENESIS IMPERFECTA (OI):

This syndrome, known as brittle bone disease, is a genetic disorder of bone fragility characterized by bones that **fracture easily**, with minor or no trauma.

Over 80% of cases of osteogenesis imperfecta (OI) are caused by dominant mutations to the genes that code for the $\alpha 1$ or $\alpha 2$ chains in type I collagen.



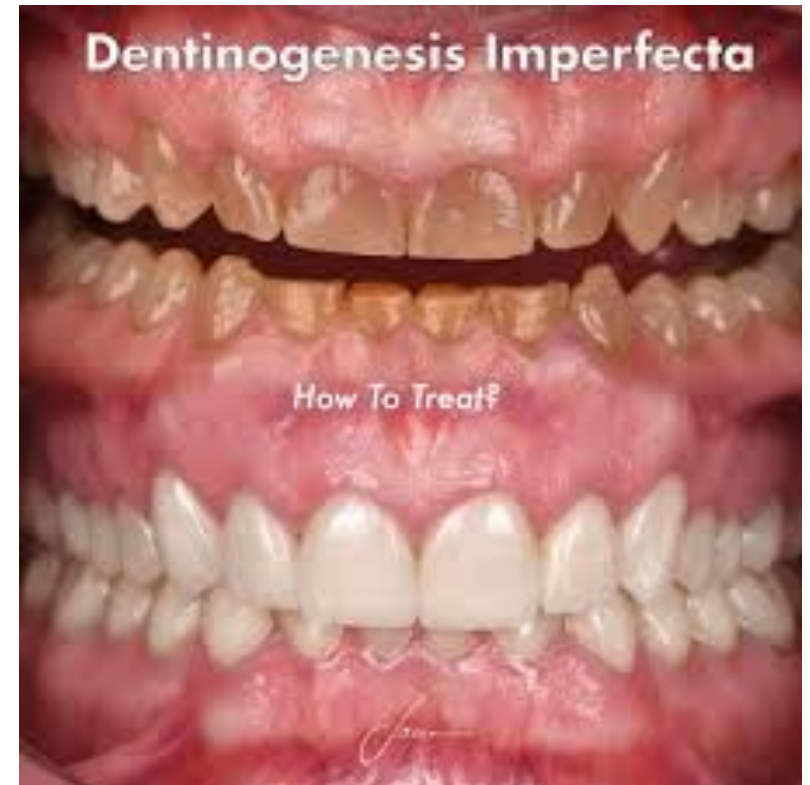
Phenotypic severity ranges from mild to lethal.

- **Type I:** the most common form, is characterized by mild bone fragility, hearing loss, and blue sclerae.
- **Type II:** the most severe form, is typically lethal in the perinatal period as a result of pulmonary complications. In utero fractures are seen.
- **Type III:** is also a severe form. It is characterized by multiple fractures at birth, short stature, spinal curvature leading to a “humped-back” (kyphotic) appearance, and blue sclerae.
- **Dentinogenesis imperfecta:** a disorder of tooth development, may be seen in OI.

DENTINOGENESIS IMPERFECTA

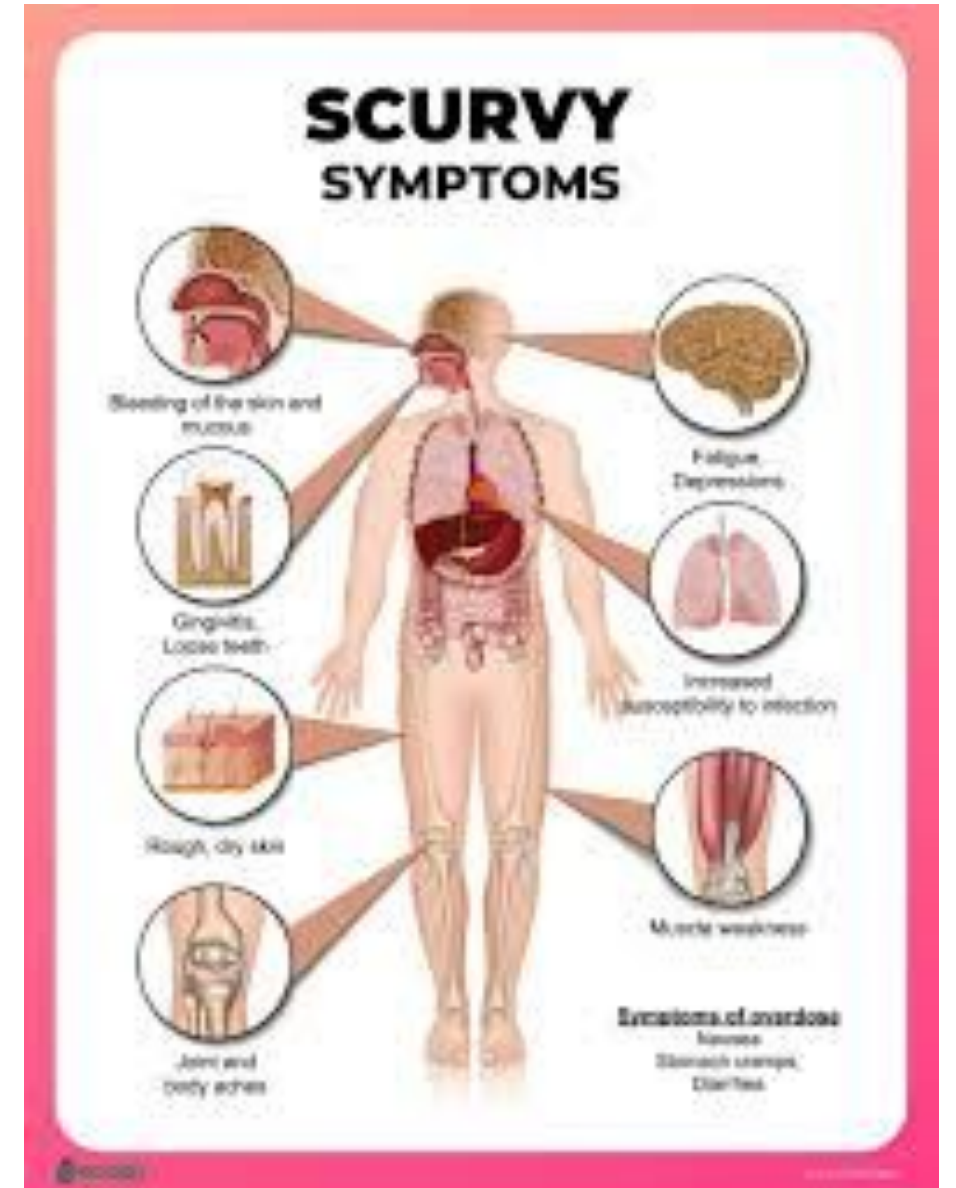
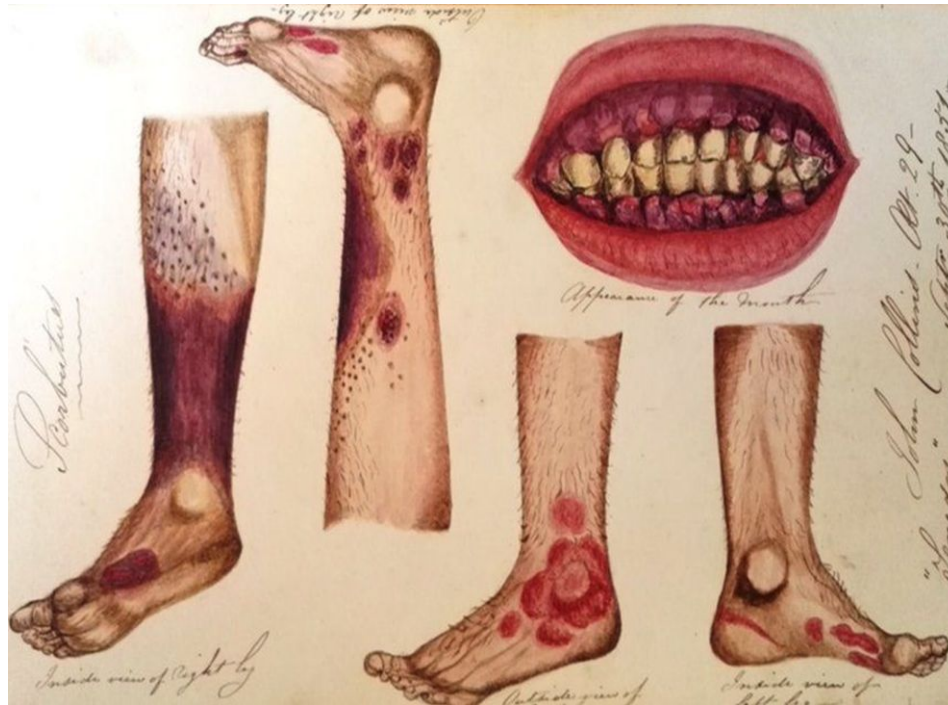
Teeth that are translucent and discolored (most often blue-grey or yellow-brown in color).

Teeth are weaker than normal, which leads to wear, breakage, and loss of teeth.



SCURVY

- The hydroxylation reactions require the reducing agent vitamin C (ascorbic acid).
- In the case of ascorbic acid deficiency, interchain formation is impaired, as is formation of a stable triple helix. Additionally, collagen fibrils cannot be cross-linked, greatly decreasing the tensile strength of the assembled fiber.
- The resulting deficiency disease is known as scurvy, a disease characterized by sore and spongy gums, loose teeth, fragile blood vessels, swollen joints, fatigue, and a microcytic anemia caused by decreased absorption of iron.
- Patients with ascorbic acid deficiency also often show bruises on the limbs as a result of subcutaneous extravasation (leakage) of blood due to capillary fragility.



ORAL SUB MUCOUS FIBROSIS(OSMF)

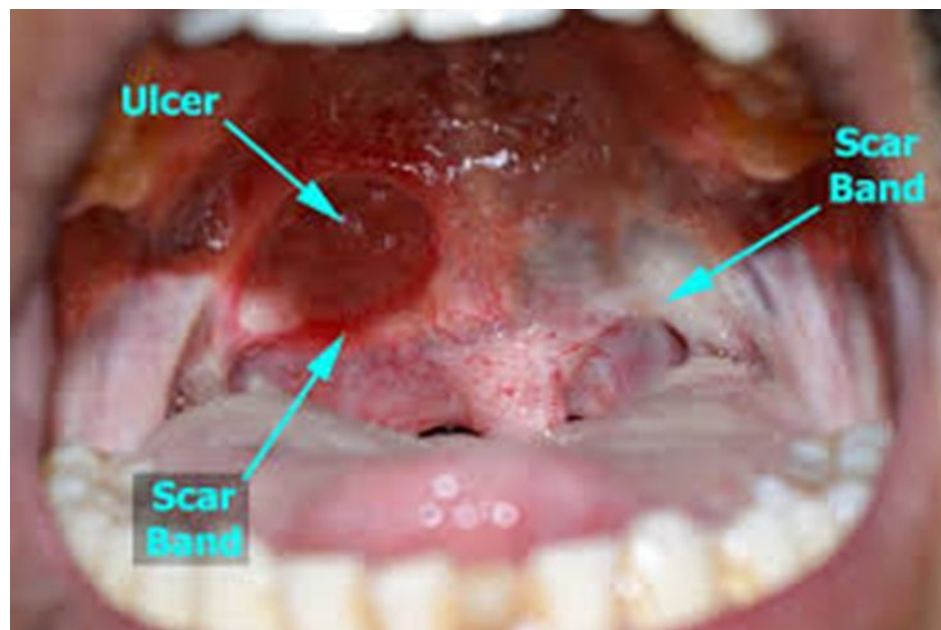
- It is a collagen metabolic disorder with an **overall increased collagen production and decreased collagen degradation**, resulting in **increased collagen deposition** in the oral tissues and fibrosis.
- Oral Sub mucous fibrosis (OSMF) has traditionally been described as “a chronic, insidious, scarring disease of the oral cavity, often with involvement of the pharynx and the upper esophagus”.
- Although the etiopathogenesis of this disease is multifactorial, areca nut-chewing in any formulation is considered the main causative agent.

CONTINUED...

- Use of areca nut and tobacco can induce fibroblast proliferation and collagen synthesis and may penetrate the oral mucosa to cause **progressive crosslinking of collagen fibers.**

The most common initial symptoms and signs are a

- ☐ burning sensation
- ☐ dry mouth
- ☐ blanching oral mucosa, and
- ☐ Ulceration
- ☐ trismus



GINGIVAL ENLARGEMENT

1. Hereditary gingival fibromatosis (HGF):

- HGF is a rare disorder characterized by a benign, non-hemorrhagic, fibrous gingival overgrowth.
- Both dominant autosomal inheritance and recessive autosomal inheritance.



2. Drug induced enlargement:

➤ Phenytoin:

- ❑ An anti epileptic drug.
- ❑ Fibroblasts become sensitive to phenytoin, and this results in subsequent increased production of collagen.
- ❑ The enzyme collagenase secreted by phenytoin-sensitive fibroblasts is relatively inactive to degrade collagen.
- ❑ An imbalance in production and degradation results in the over accumulation of collagen and hence in an increase in the bulk of connective tissue.

❑ Calcium channel blockers:

After an interaction between nifedipine and gingival fibroblasts, **overproduction of collagen and extracellular ground substance occurs** and leads to an increase in the size of the gingiva.

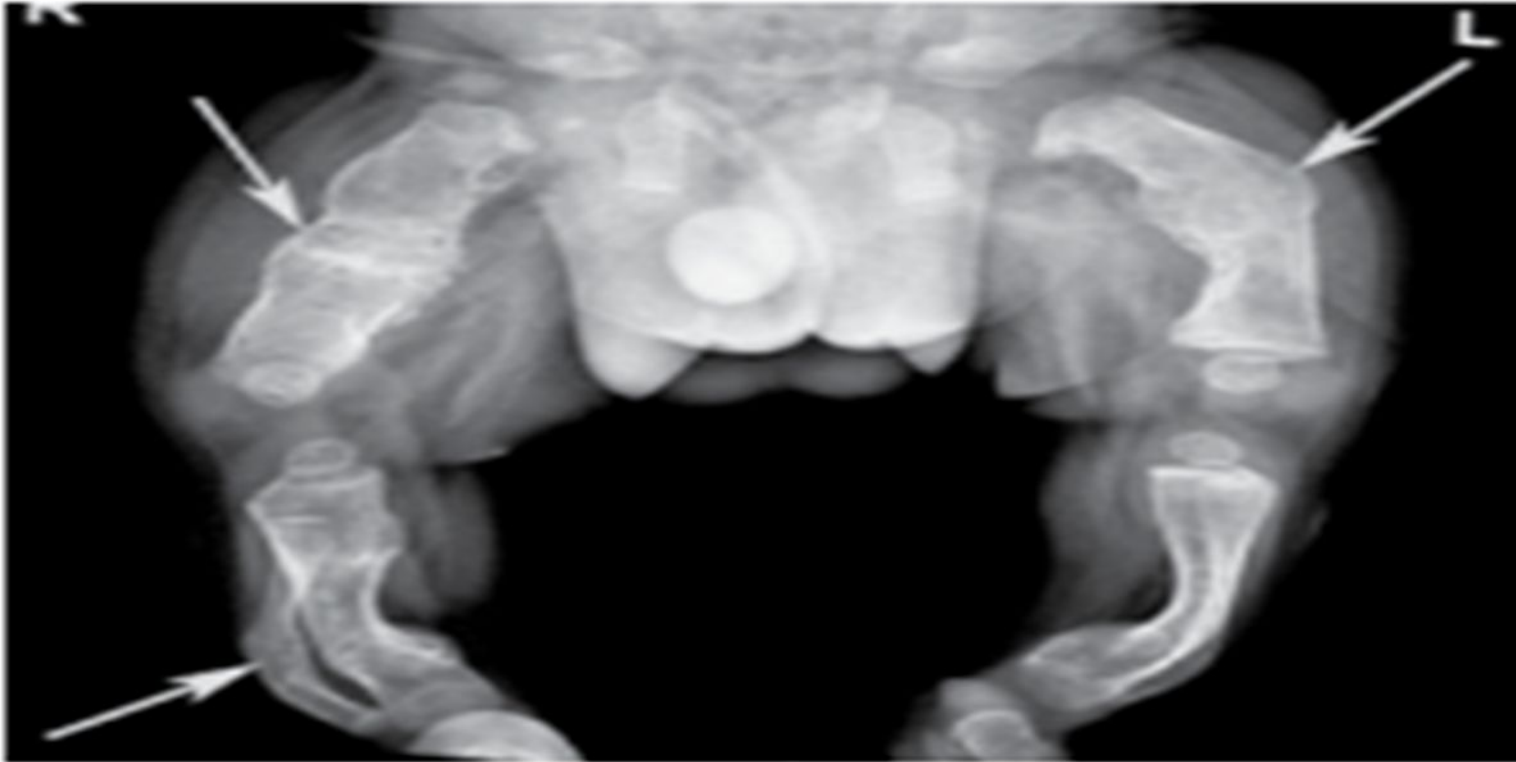
❑ Cyclosporin:

It could react with a phenotypically distinct subpopulation of **gingival fibroblasts to enhance protein synthesis.**



SCENARIO 1

- A 2-month-old male baby was brought to the Orthopedic Outpatient Department with complaints of **deformities of both lower limbs following birth**.
- There was no history of trauma and similar complaints in the rest of the 2 older siblings. The baby appeared **abnormal with bilateral bowed (curved) lower limbs**. However, his **serum alkaline phosphatase** was slightly high. The x-ray of the lower limbs is shown in Figure



An x-ray of both lower limbs - anteroposterior view showing **multiple fractures** in the long bones of the lower limbs at different stages of healing, with **excessive callus formation** (arrows). Radiograph also showed **flaring and bowing of the ends of the long bones**.

What is the diagnosis?

OSTEOGENESIS IMPERFECTA (OI)

- the presence of multiple fractures of the long bones
- flaring and bowing of the ends of the long bones

What are the associated abnormalities?

- hearing loss
- blue sclera

Dentinogenesis imperfecta are the associated abnormalities, which one can encounter in patients.



Scenario 2

A 10-year-old boy reported to our college with chief complaint of inability in chewing caused by missing teeth since past 2 years.

Patient has had **joint hypermobility history since childhood.**

Physical examination revealed **hyperelasticity of the skin.**

On oral examination, **absence of the inferior labial and lingual frenula** has been reported.

What is the diagnosis?

- **EHLER DANLOS SYNDROME**
 - Joint hypermobility
 - Hyperelasticity of the skin
 - Absence of inferior labial and lingual frenula
 - Premature exfoliation of posterior deciduous teeth within five years and unerupted permanent teeth.

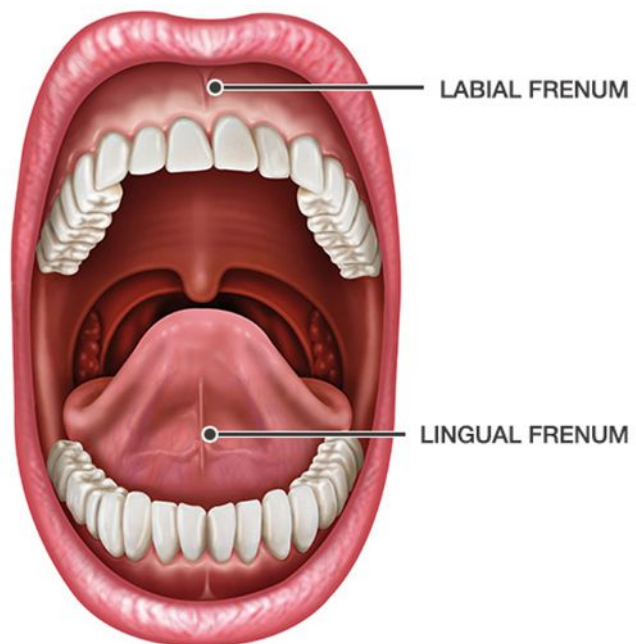
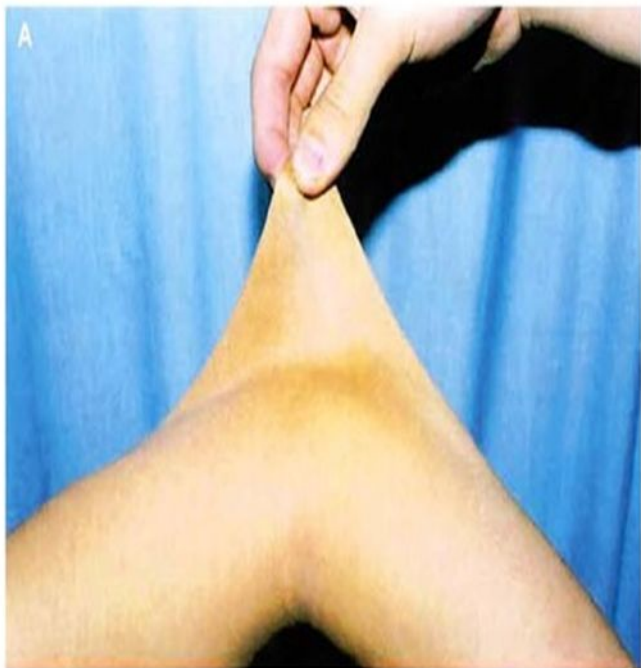


Fig. 2. Absent lingual frenulum.

Scenario 3

- A 32-year-old male patient reported to the department with a **burning sensation in the mouth**.
- He did not provide any significant medical history.
- On intraoral examination, **fibrous bands were palpable** bilaterally and **tongue protrusion was significantly reduced**.
- His mouth opening was 26 mm.
- Diagnosis: Oral sub mucous fibrosis

Scenario 4

- 4 year old male pt was brought in for consultation exhibiting possible anemia, edema in his right leg, bleeding gingiva (signs of tooth loss visible), irritability, paleness, sweating, dry/cracked hair, and apparent signs of malnourishment.
- Patient is afebrile with no signs of organomegaly.
- Pt is the 10th born child of a low income household. This family has experienced infant mortality previously in the form of malnutrition and dehydration.

What is the diagnosis?

- **SCURVY**

Deficiency of vitamin C

Nonhydroxylated chain is not able to mature



The stable triple helix cannot be formed



Immediate degradation inside the cell



Loss of collagen in the matrix



Falling out of teeth
Vascular fragility
Poor wound healing

DEFECTS OF ELASTINS

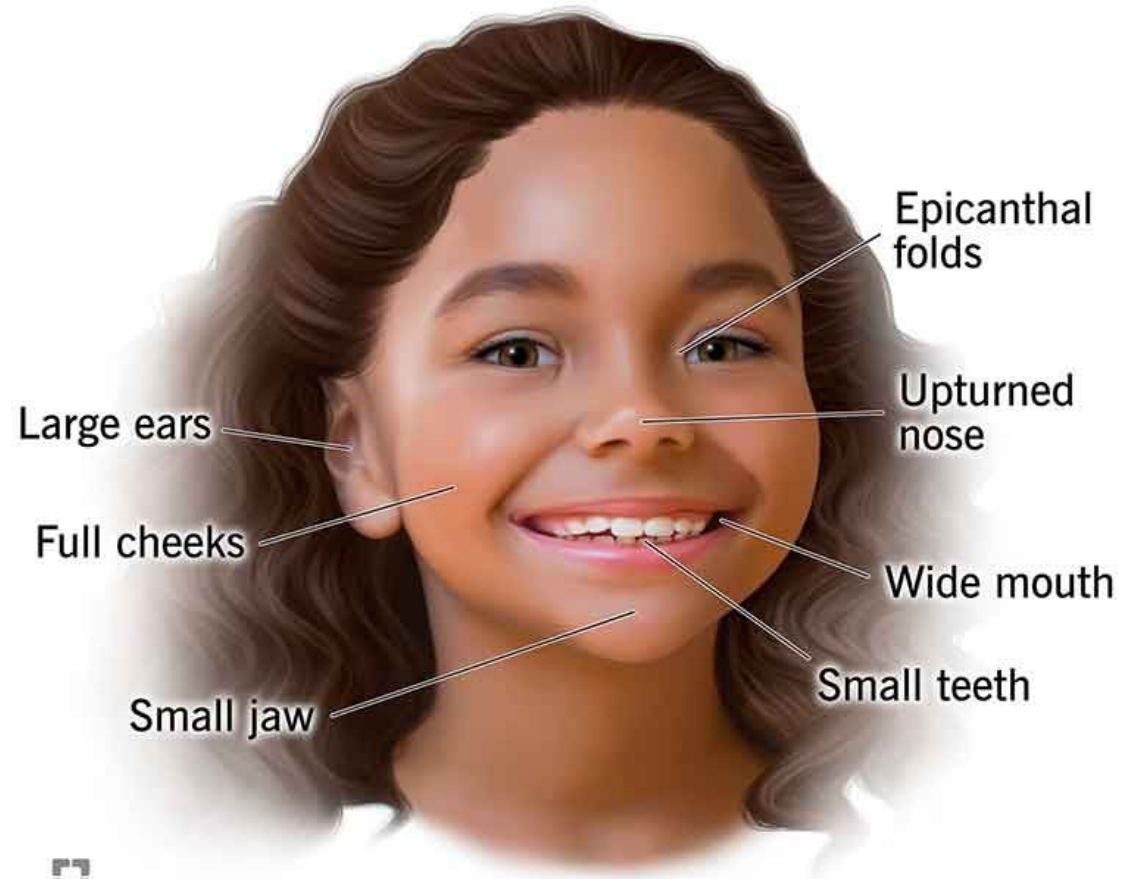
In biochemistry, elastin defects are primarily categorized into **genetic mutations** (affecting the *ELN* gene) and **acquired degradation** (often involving an imbalance of proteases).

1. WILLIAMS-BEUREN SYNDROME (WBS)

This is a multisystem disorder caused by a **microdeletion** on chromosome 7q11.23, which includes the *ELN* (elastin) gene.

- **Biochemical Impact:** Hemizygosity (having only one functional copy of the gene) leads to **elastin insufficiency**.
- **Clinical Presentation:** * **Supravalvular Aortic Stenosis (SVAS):** Narrowing of the aorta due to disorganized elastic fibers in the vessel wall.
 - **Connective Tissue Laxity:** Leads to a distinctive "elfin" facial appearance and joint hypermobility.

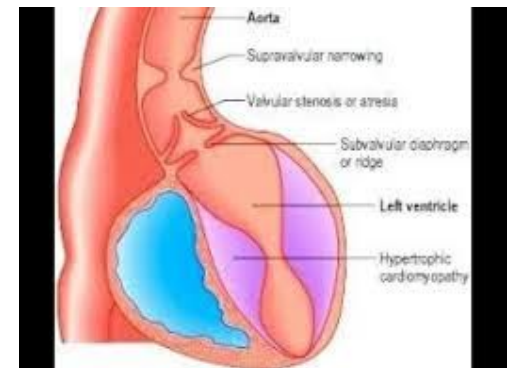
Williams syndrome



2. SUPRAVALVULAR AORTIC STENOSIS (SVAS)

While SVAS is part of Williams Syndrome, it can also occur as an **autosomal dominant** isolated defect.

- **Mechanism:**
- Mutations in the *ELN* gene result in truncated elastin proteins. This triggers an over-proliferation of smooth muscle cells in the arterial wall as a compensatory mechanism, eventually narrowing the lumen of the aorta.

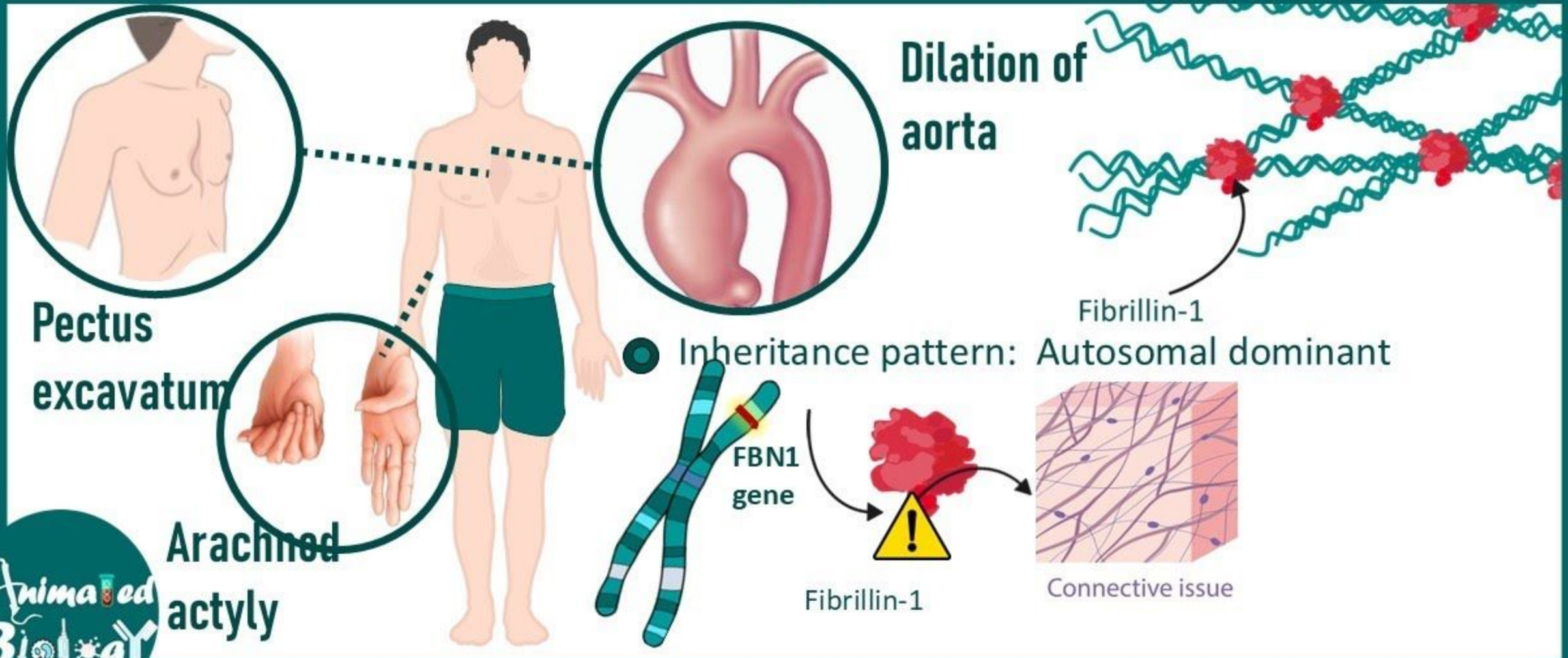


3. MARFAN SYNDROME (THE FIBRILLIN CONNECTION)

Marfan Syndrome is a defect in **Fibrillin-1**, not the elastin molecule itself. However, because Fibrillin-1 acts as the mandatory **scaffold** for elastin deposition, the elastin cannot function.

- **Biochemical Impact:** Elastic lamellae in the large arteries are fragmented and disorganized (**cystic medial necrosis**).
- **Clinical Danger:** * **Aortic Aneurysm and Dissection:** The aorta loses its ability to recoil, stretching under pressure until it tears.
 - **Ectopia Lentis:** Dislocation of the lens of the eye due to weak suspensory ligaments (which are rich in fibrillin).

Marfan syndrome



4. ALPHA-1 ANTITRYPSIN (A1AT) DEFICIENCY

This is the classic biochemical example of an **acquired** elastin defect.

The Enzyme Balance: In healthy lungs, **Neutrophil Elastase** (which breaks down elastin) is kept in check by **A1AT** (a protease inhibitor).

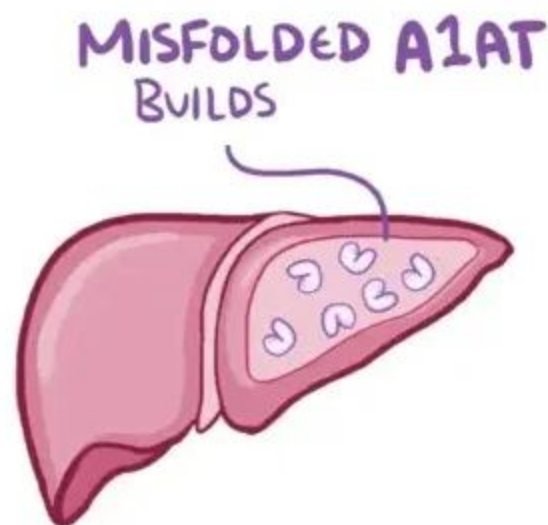
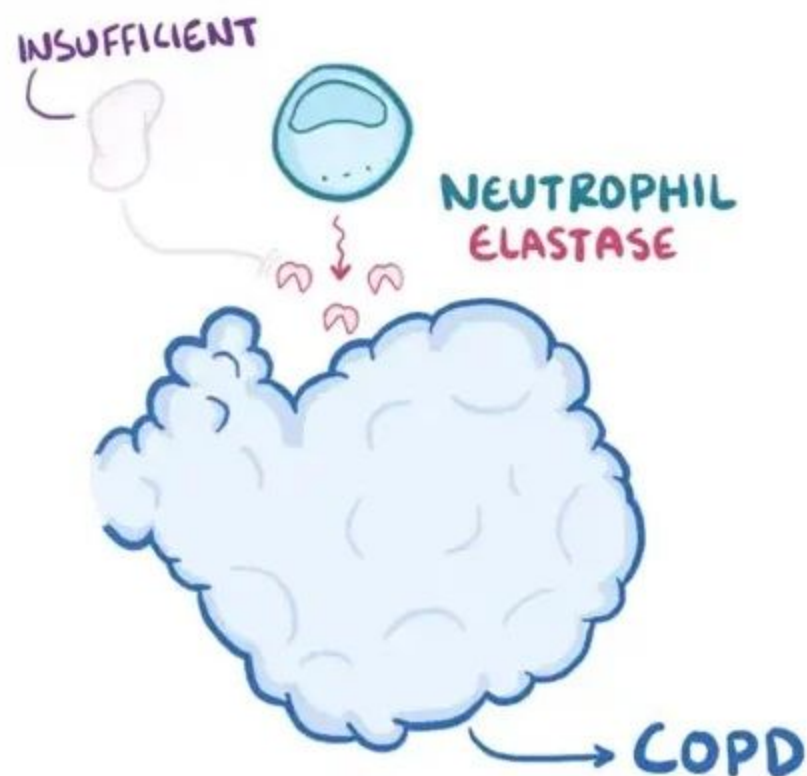
Pathology: In A1AT deficiency, the inhibitor is missing or defective. Neutrophil elastase goes unchecked and "digests" the elastic fibers in the alveolar walls of the lungs.

Clinical Result: Emphysema. The lungs lose their elastic recoil; they can expand but cannot easily push air out, leading to chronic obstructive pulmonary disease (COPD).

Alpha-1-Antitrypsin Test Purpose, Procedure, Result and more

web: www.labtestsguide.com | Email: info@labtestsguide.com

ALPHA-1 ANTITRYPSIN DEFICIENCY



Also Known as

1. A1AT
2. AAT
3. Alpha-1-Antitrypsin Deficiency
4. Alpha-1-Antitrypsin



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5. CUTIS LAXA (ELASTOLYSIS)

This is a heterogeneous group of disorders where elastic fibers are either absent, sparse, or fragmented.

Etiology: Can be inherited (mutations in *ELN* or *FBLN5*) or acquired (often following inflammatory skin diseases).

Clinical Presentation: The skin becomes inelastic and hangs in loose, pendulous folds, giving even young children a prematurely aged appearance. Unlike Ehlers-Danlos, the skin is **not** hyper-extensible (it doesn't snap back; it just hangs).



A



B



C

PSEUDOXANTHOMA ELASTICUM (PXE)

A metabolic disorder caused by mutations in the *ABCC6* gene.

- **Biochemical Impact:** Abnormal **mineralization (calcification)** of elastic fibers.
- **Clinical Features:** * **Skin:** "Plucked chicken" appearance (yellowish papules).
 - **Eyes:** Angioid streaks in the retina.
 - **Vascular:** Premature atherosclerosis and intermittent claudication due to calcified arteries.

Pseudoxanthoma Elasticum



Calcified tissue becomes firmer and less stretchy



Angioid streaks

Active lesion

Subretinal fibrosis

Feature

Collagen Defect (e.g., EDS)

Elastin Defect (e.g., SVAS/A1AT)

Primary Loss

Tensile Strength (Tear resistance)

Elastic Recoil (Snap-back ability)

Key Symptom

Fragility, hyper-mobility

Aneurysms, obstructive lung disease

Common Enzyme

Lysyl Oxidase (Cross-linking)

Elastase (Degradation)

